

## Lab 3 - Routing Protocol

NAME: LIM YU HAN

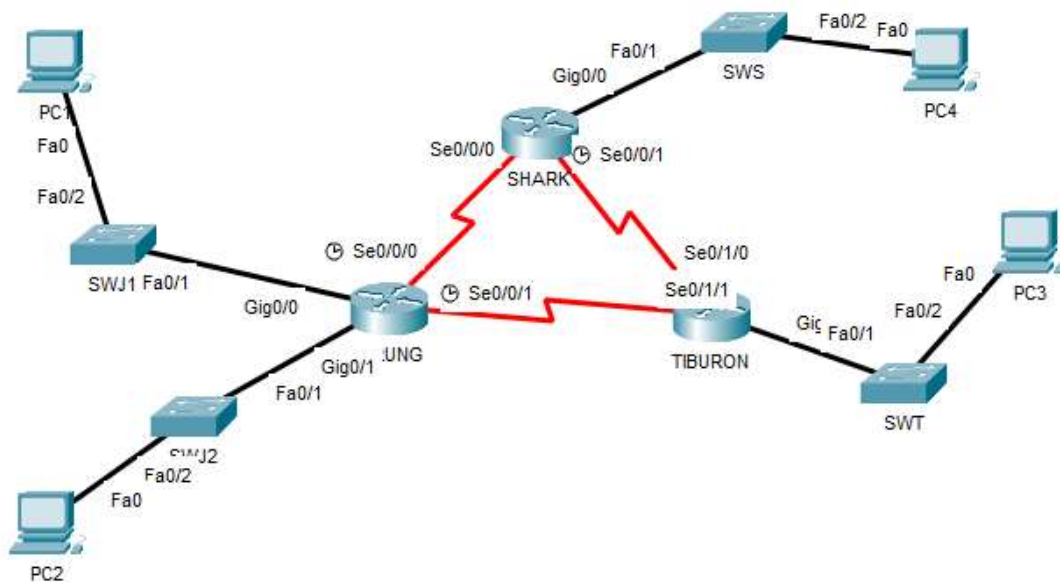
MATRIC NUM: A23CS0241

SECTION: 12

### INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

### TOPOLOGY



## LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

**Table 1**

| Device      | Subnetwork     | Usable Hosts |
|-------------|----------------|--------------|
| JERUNG      | LAN1           | 20           |
|             | LAN2           | 30           |
| SHARK       | LAN            | 60           |
| TIBURON     | LAN            | 50           |
| Connections | JERUNG-SHARK   | 2            |
|             | JERUNG-TIBURON | 2            |
|             | SHARK-TIBURON  | 2            |

**Table 2**

| #  | Device Name | Interface | IP Address      | Subnet Mask     | Gateway        |
|----|-------------|-----------|-----------------|-----------------|----------------|
| 1  | JERUNG      | Se0/0/0   | 172.18.110.193  | 255.255.255.252 | -              |
| 2  |             | Se0/0/1   | 172.18.110.197  | 255.255.255.252 | -              |
| 3  |             | G0/0      | 172.18.110.1    | 255.255.255.224 | -              |
| 4  |             | G0/1      | 172.18.110.33   | 255.255.255.224 | -              |
| 5  | TIBURON     | Se0/1/1   | 172.18.110.201  | 255.255.255.252 | -              |
| 6  |             | Se0/1/0   | 172.18.110.198  | 255.255.255.252 | -              |
| 7  |             | G0/0      | 172.168.110.129 | 255.255.255.192 | -              |
| 8  | SHARK       | Se0/0/0   | 172.18.110.194  | 255.255.255.252 | -              |
| 9  |             | Se0/0/1   | 172.18.110.201  | 255.255.255.252 | -              |
| 10 |             | G0/0      | 172.18.110.65   | 255.255.255.192 | -              |
| 11 | PC1         | -         | 172.18.110.30   | 255.255.255.224 | 172.18.110.1   |
| 12 | PC2         | -         | 172.18.110.62   | 255.255.255.224 | 172.18.110.33  |
| 13 | PC3         | -         | 172.18.110.190  | 255.255.255.192 | 172.18.110.129 |
| 14 | PC4         | -         | 172.18.110.126  | 255.255.255.192 | 172.18.110.65  |

## LAB TASKS

### Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
- Using the IP addresses calculated, configure the devices with the appropriate information.

| Device      | Subnetwork     | Usable Hosts | Total IPs  | Usable IPs | Prefix         |
|-------------|----------------|--------------|------------|------------|----------------|
| JERUNG      | LAN1           | 20           | $2^5 = 32$ | 30         | $32 - 5 = /27$ |
|             | LAN2           | 30           | $2^5 = 32$ | 30         | $32 - 5 = /27$ |
| SHARK       | LAN            | 60           | $2^6 = 64$ | 62         | $32 - 6 = /26$ |
| TIBURON     | LAN            | 50           | $2^6 = 64$ | 62         | $32 - 6 = /26$ |
| Connections | JERUNG-SHARK   | 2            | $2^5 = 32$ | 30         | $32 - 2 = /30$ |
|             | JERUNG-TIBURON | 2            | $2^2 = 4$  | 2          | $32 - 2 = /30$ |
|             | SHARK-TIBURON  | 2            | $2^2 = 4$  | 2          | $32 - 2 = /30$ |

| Device      | Subnetwork     | Total IPs                       | Usable IPs                       |
|-------------|----------------|---------------------------------|----------------------------------|
| JERUNG      | LAN1           | 172.18.110.0 – 172.18.110.31    | 172.18.110.1 – 172.18.0.30       |
|             | LAN2           | 172.18.110.32 – 172.18.110.63   | 172.18.110.33 – 172.18.0.62      |
| SHARK       | LAN            | 172.18.110.64 – 172.18.110.127  | 172.18.110.65 – 172.18.110.126   |
| TIBURON     | LAN            | 172.18.110.128 – 172.18.110.191 | 172.168.110.129 – 172.18.110.190 |
| Connections | JERUNG-SHARK   | 172.18.110.192 – 172.18.110.195 | 172.18.110.193 – 172.18.110.194  |
|             | JERUNG-TIBURON | 172.18.110.196 – 172.18.110.199 | 172.18.110.197 – 172.18.110.198  |
|             | SHARK-TIBURON  | 172.18.110.200 – 172.18.110.203 | 172.18.110.201 – 172.18.110.202  |

| Device      | Subnetwork     | Prefix | Subnet Mask                                           |
|-------------|----------------|--------|-------------------------------------------------------|
| JERUNG      | LAN1           | /27    | 11111111.11111111.11111111.11100000 = 255.255.255.224 |
|             | LAN2           | /27    | 11111111.11111111.11111111.11111100 = 255.255.255.252 |
| SHARK       | LAN            | /26    | 11111111.11111111.11111111.11000000 = 255.255.255.192 |
| TIBURON     | LAN            | /26    | 11111111.11111111.11111111.11000000 = 255.255.255.192 |
| Connections | JERUNG-SHARK   | /30    | 11111111.11111111.11111111.11111100 = 255.255.255.252 |
|             | JERUNG-TIBURON | /30    | 11111111.11111111.11111111.11111100 = 255.255.255.252 |
|             | SHARK-TIBURON  | /30    | 11111111.11111111.11111111.11111100 = 255.255.255.252 |

### Task 2 – Routing Table

- Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
  - CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

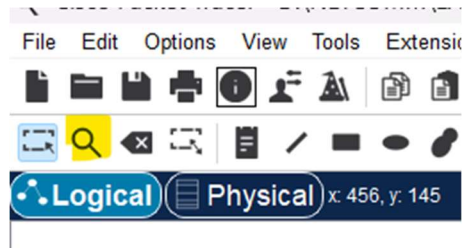


Figure B

| Type | Network         | Port               | Next Hop IP | Metric |
|------|-----------------|--------------------|-------------|--------|
| C    | 172.18.110.0/26 | GigabitEthernet0/0 | ---         | 0/0    |
| L    | 172.18.110.1/32 | GigabitEthernet0/0 | ---         | 0/0    |

Figure C

## JERUNG:

### CLI

```

JERUNG>enable
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

```

### PT TOOL

| Routing Table for JERUNG |                   |                    |             |        |
|--------------------------|-------------------|--------------------|-------------|--------|
| Type                     | Network           | Port               | Next Hop IP | Metric |
| C                        | 172.18.110.0/27   | GigabitEthernet0/0 | ---         | 0/0    |
| L                        | 172.18.110.1/32   | GigabitEthernet0/0 | ---         | 0/0    |
| C                        | 172.18.110.32/27  | GigabitEthernet0/1 | ---         | 0/0    |
| L                        | 172.18.110.33/32  | GigabitEthernet0/1 | ---         | 0/0    |
| C                        | 172.18.110.192/30 | Serial0/0/0        | ---         | 0/0    |
| L                        | 172.18.110.193/32 | Serial0/0/0        | ---         | 0/0    |
| C                        | 172.18.110.196/30 | Serial0/0/1        | ---         | 0/0    |
| L                        | 172.18.110.197/32 | Serial0/0/1        | ---         | 0/0    |

## TIBURON:

### CLI

```
TIBURON>enable
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.18.110.196/30 is directly connected, Serial0/1/0
L       172.18.110.198/32 is directly connected, Serial0/1/0
C       172.18.110.200/30 is directly connected, Serial0/1/1
L       172.18.110.201/32 is directly connected, Serial0/1/1
    172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.168.110.128/26 is directly connected, GigabitEthernet0/0
L       172.168.110.129/32 is directly connected, GigabitEthernet0/0
```

### PT TOOL

Routing Table for TIBURON

| Type | Network            | Port               | Next Hop IP | Metric |
|------|--------------------|--------------------|-------------|--------|
| C    | 172.18.110.196/30  | Serial0/1/0        | ---         | 0/0    |
| L    | 172.18.110.198/32  | Serial0/1/0        | ---         | 0/0    |
| C    | 172.18.110.200/30  | Serial0/1/1        | ---         | 0/0    |
| L    | 172.18.110.201/32  | Serial0/1/1        | ---         | 0/0    |
| C    | 172.168.110.128/26 | GigabitEthernet0/0 | ---         | 0/0    |
| L    | 172.168.110.129/32 | GigabitEthernet0/0 | ---         | 0/0    |

## SHARK:

### CLI

```

SHARK>enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

```

### PT TOOL

Routing Table for SHARK

| Type | Network           | Port               | Next Hop IP | Metric |
|------|-------------------|--------------------|-------------|--------|
| C    | 172.18.110.64/26  | GigabitEthernet0/0 | ---         | 0/0    |
| L    | 172.18.110.65/32  | GigabitEthernet0/0 | ---         | 0/0    |
| C    | 172.18.110.192/30 | Serial0/0/0        | ---         | 0/0    |
| L    | 172.18.110.194/32 | Serial0/0/0        | ---         | 0/0    |
| C    | 172.18.110.200/30 | Serial0/0/1        | ---         | 0/0    |
| L    | 172.18.110.201/32 | Serial0/0/1        | ---         | 0/0    |

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=12ms TTL=128
Reply from 172.18.110.62: bytes=32 time=8ms TTL=128
Reply from 172.18.110.62: bytes=32 time<1ms TTL=128
Reply from 172.18.110.62: bytes=32 time=2ms TTL=128

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 5ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

-PC1 and PC2 ping successfully because they are on the same subnet.  
-PC1 and PC4 are unable to communicate directly because they are on different subnets and connected to different routers. As a result, PC1 cannot determine the path to PC4, leading to the "Destination host unreachable" message. Additionally, PC4 has an incorrectly configured default gateway, preventing it from sending replies back to PC1. These combined issues including different subnets, separate routers, and misconfigured gateway settings, result in failed communication between the two devices.

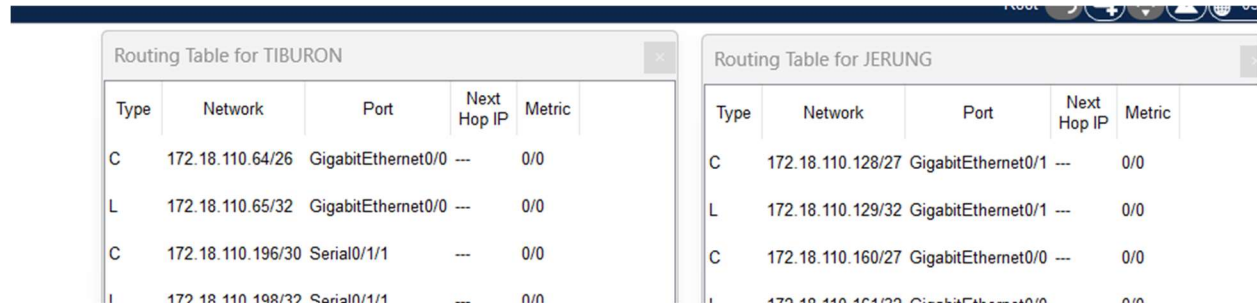
5. What needs to be done to ensure all PCs can ping each other successfully?

- All interfaces should have correct IP addresses and subnet masks.
- Default gateways must be properly configured on all PCs to ensure packet forwarding.
- Routing information must be shared between routers using RIP or static routes.
- Ensure no access control lists (ACLs) are blocking communication.



## Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.



| Type | Network           | Port               | Next Hop IP | Metric |
|------|-------------------|--------------------|-------------|--------|
| C    | 172.18.110.64/26  | GigabitEthernet0/0 | ---         | 0/0    |
| L    | 172.18.110.65/32  | GigabitEthernet0/0 | ---         | 0/0    |
| C    | 172.18.110.196/30 | Serial0/1/1        | ---         | 0/0    |
| L    | 172.18.110.198/32 | Serial0/1/1        | ---         | 0/0    |

| Type | Network           | Port               | Next Hop IP | Metric |
|------|-------------------|--------------------|-------------|--------|
| C    | 172.18.110.128/27 | GigabitEthernet0/1 | ---         | 0/0    |
| L    | 172.18.110.129/32 | GigabitEthernet0/1 | ---         | 0/0    |
| C    | 172.18.110.160/27 | GigabitEthernet0/0 | ---         | 0/0    |
| L    | 172.18.110.161/32 | GigabitEthernet0/0 | ---         | 0/0    |

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- What can you say about the addresses used in the 'network' instructions in Figure E?

The network' instructions are used to tell RIP to advertise and include all interface within the subnets.

network 172.18.110.128 – include all interface within 172.18.110.128 subnet

network 172.18.110.160 – include all interface within 172.18.110.160 subnet

network 172.18.110.192 – include all interface within 172.18.110.192 subnet

network 172.18.110.196 – include all interface within 172.18.110.196 subnet

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

### JERUNG

| Type | Network           | Port                   | Next Hop IP    | Metric |
|------|-------------------|------------------------|----------------|--------|
| C    | 172.18.110.0/27   | GigabitEthernet0/0 --- |                | 0/0    |
| L    | 172.18.110.1/32   | GigabitEthernet0/0 --- |                | 0/0    |
| C    | 172.18.110.32/27  | GigabitEthernet0/1 --- |                | 0/0    |
| L    | 172.18.110.33/32  | GigabitEthernet0/1 --- |                | 0/0    |
| R    | 172.18.110.64/26  | Serial0/0/0            | 172.18.110.194 | 120/1  |
| R    | 172.18.110.128/26 | Serial0/0/1            | 172.18.110.198 | 120/1  |
| C    | 172.18.110.192/30 | Serial0/0/0            | ---            | 0/0    |
| L    | 172.18.110.193/32 | Serial0/0/0            | ---            | 0/0    |
| C    | 172.18.110.196/30 | Serial0/0/1            | ---            | 0/0    |
| L    | 172.18.110.197/32 | Serial0/0/1            | ---            | 0/0    |
| R    | 172.18.110.200/30 | Serial0/0/1            | 172.18.110.198 | 120/1  |
| R    | 172.18.110.200/30 | Serial0/0/0            | 172.18.110.194 | 120/1  |

## TIBURON

| Type | Network           | Port               | Next Hop IP    | Metric |
|------|-------------------|--------------------|----------------|--------|
| R    | 172.18.110.0/27   | Serial0/1/1        | 172.18.110.197 | 120/1  |
| R    | 172.18.110.32/27  | Serial0/1/1        | 172.18.110.197 | 120/1  |
| R    | 172.18.110.64/26  | Serial0/1/0        | 172.18.110.201 | 120/1  |
| C    | 172.18.110.128/26 | GigabitEthernet0/0 | ---            | 0/0    |
| L    | 172.18.110.129/32 | GigabitEthernet0/0 | ---            | 0/0    |
| R    | 172.18.110.192/30 | Serial0/1/1        | 172.18.110.197 | 120/1  |
| R    | 172.18.110.192/30 | Serial0/1/0        | 172.18.110.201 | 120/1  |
| C    | 172.18.110.196/30 | Serial0/1/1        | ---            | 0/0    |
| L    | 172.18.110.198/32 | Serial0/1/1        | ---            | 0/0    |
| C    | 172.18.110.200/30 | Serial0/1/0        | ---            | 0/0    |
| L    | 172.18.110.202/32 | Serial0/1/0        | ---            | 0/0    |

- a. What are the changes seen in TIBURON?

Before RIP, the routing table only showed directly connected networks. After RIP, TIBURON's table includes RIP-learned routes (R) for the networks connected to JERUNG and SHARK.

- b. What are the Networks with type R in TIBURON and JERUNG?

TIBURON:

- 172.18.110.0/27 (LAN1 on JERUNG)
- 172.18.110.32/27 (LAN2 on JERUNG)
- 172.18.110.64/26 (SHARK LAN)

JERUNG:

- 172.18.110.64/26 (SHARK LAN)
- 172.18.110.128/26 (TIBURON LAN)

- c. Ping PC3 from PC1. Was it successful?

YES.

```

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=3ms TTL=128
Reply from 172.18.110.190: bytes=32 time=2ms TTL=128
Reply from 172.18.110.190: bytes=32 time=3ms TTL=128
Reply from 172.18.110.190: bytes=32 time=1ms TTL=128

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

```

- d. Ping PC4 from PC1. Was it successful?

NO.

```

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

- e. Explain the reasons for your answer.

- PC1 and PC3 ping successfully because they are on the same subnet.
- PC1 and PC4 are on different subnet. They cannot communicate directly because they are in different network, and connected to different routers. The router needs to forward the packets between the network.
- PC1 does not know the path for PC4, and it shows Destination host unreachable.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```

SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.64
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary

```

- g. Open router SHARK's routing table and paste here.

```

R    172.18.110.0/27    Serial0/0/0          172.18.110.193 120/1
R    172.18.110.32/27   Serial0/0/0          172.18.110.193 120/1
C    172.18.110.64/26   GigabitEthernet0/0   ---            0/0
L    172.18.110.65/32   GigabitEthernet0/0   ---            0/0
R    172.18.110.128/26  Serial0/0/1          172.18.110.202 120/1
C    172.18.110.192/30  Serial0/0/0          ---            0/0
L    172.18.110.194/32  Serial0/0/0          ---            0/0
R    172.18.110.196/30  Serial0/0/0          172.18.110.193 120/1
R    172.18.110.196/30  Serial0/0/1          172.18.110.202 120/1
C    172.18.110.200/30  Serial0/0/1          ---            0/0
L    172.18.110.201/32  Serial0/0/1          ---            0/0

```

- h. Try to ping from PC4 to all other PCs in the topology. \*Note: Try to ping at least twice to get best results.

| PC | Ping result |
|----|-------------|
| 1  | Success     |
| 2  | Success     |
| 3  | Success     |

## PC1

```
C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=3ms TTL=128
Reply from 172.18.110.30: bytes=32 time<1ms TTL=128
Reply from 172.18.110.30: bytes=32 time<1ms TTL=128
Reply from 172.18.110.30: bytes=32 time=11ms TTL=128

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=3ms TTL=128
Reply from 172.18.110.30: bytes=32 time=3ms TTL=128
Reply from 172.18.110.30: bytes=32 time=13ms TTL=128
Reply from 172.18.110.30: bytes=32 time=3ms TTL=128

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 13ms, Average = 5ms
```

## PC2

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=4ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128
Reply from 172.18.110.62: bytes=32 time<1ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=4ms TTL=128
Reply from 172.18.110.62: bytes=32 time=4ms TTL=128
Reply from 172.18.110.62: bytes=32 time<1ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 2ms
```



## PC3

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time<1ms TTL=128
Reply from 172.18.110.190: bytes=32 time=10ms TTL=128
Reply from 172.18.110.190: bytes=32 time=3ms TTL=128
Reply from 172.18.110.190: bytes=32 time=11ms TTL=128

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 6ms

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=3ms TTL=128
Reply from 172.18.110.190: bytes=32 time=5ms TTL=128
Reply from 172.18.110.190: bytes=32 time<1ms TTL=128
Reply from 172.18.110.190: bytes=32 time=2ms TTL=128

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 2ms
```

### Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
  - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0/24](#)
  - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

| PC3             | Info                          |
|-----------------|-------------------------------|
| IP address      | <a href="#">192.168.1.2</a>   |
| Subnet Mask     | <a href="#">255.255.255.0</a> |
| Gateway Address | <a href="#">192.168.1.1</a>   |

- c. After this change, can PC4 and PC1 ping PC3?  
[No, because they are on different gateway.](#)

[Ping PC3 from PC4](#)

```
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

#### Ping PC3 from PC1

```
Pinging 198.162.1.2 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 198.162.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

#### SHARK

|   |                   |                    |                |       |
|---|-------------------|--------------------|----------------|-------|
| R | 172.18.110.0/27   | Serial0/0/0        | 172.18.110.193 | 120/1 |
| R | 172.18.110.32/27  | Serial0/0/0        | 172.18.110.193 | 120/1 |
| C | 172.18.110.64/26  | GigabitEthernet0/0 | ---            | 0/0   |
| L | 172.18.110.65/32  | GigabitEthernet0/0 | ---            | 0/0   |
| C | 172.18.110.192/30 | Serial0/0/0        | ---            | 0/0   |
| L | 172.18.110.194/32 | Serial0/0/0        | ---            | 0/0   |
| R | 172.18.110.196/30 | Serial0/0/0        | 172.18.110.193 | 120/1 |
| R | 172.18.110.196/30 | Serial0/0/1        | 172.18.110.202 | 120/1 |
| C | 172.18.110.200/30 | Serial0/0/1        | ---            | 0/0   |
| L | 172.18.110.201/32 | Serial0/0/1        | ---            | 0/0   |



## TIBURON

|   |                   |                    |                |       |
|---|-------------------|--------------------|----------------|-------|
| R | 172.18.110.0/27   | Serial0/1/1        | 172.18.110.197 | 120/1 |
| R | 172.18.110.32/27  | Serial0/1/1        | 172.18.110.197 | 120/1 |
| R | 172.18.110.64/26  | Serial0/1/0        | 172.18.110.201 | 120/1 |
| R | 172.18.110.192/30 | Serial0/1/1        | 172.18.110.197 | 120/1 |
| R | 172.18.110.192/30 | Serial0/1/0        | 172.18.110.201 | 120/1 |
| C | 172.18.110.196/30 | Serial0/1/1        | ---            | 0/0   |
| L | 172.18.110.198/32 | Serial0/1/1        | ---            | 0/0   |
| C | 172.18.110.200/30 | Serial0/1/0        | ---            | 0/0   |
| L | 172.18.110.202/32 | Serial0/1/0        | ---            | 0/0   |
| C | 192.168.1.0/26    | GigabitEthernet0/0 | ---            | 0/0   |
| L | 192.168.1.1/32    | GigabitEthernet0/0 | ---            | 0/0   |

- e. Referring to the routing table, explain your findings.

-TIBURON is directly connected to 192.168.1.0/24 subnet and route traffic for other subnets, including those in the 172.18.110.0/24.

-SHARK is directly connected to 172.18.110.64/26, and has routing information for 172.18.110.0/27 and 172.18.110.32/27 subnets.

-Both routers rely on RIP, which shows R (RIP-learn) in the routing table.

- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

Ensure all routers have the correct network statements in RIP to propagate 192.168.1.0/24 by updating RIP on TIBURON and propagate the new route for other subnets.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

| PC | Ping result |
|----|-------------|
| 1  | Success     |
| 2  | Success     |
| 4  | Success     |

## PC1

```
C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=5ms TTL=128
Reply from 172.18.110.30: bytes=32 time=14ms TTL=128
Reply from 172.18.110.30: bytes=32 time=3ms TTL=128
Reply from 172.18.110.30: bytes=32 time=3ms TTL=128

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 14ms, Average = 6ms

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=4ms TTL=128
Reply from 172.18.110.30: bytes=32 time=3ms TTL=128
Reply from 172.18.110.30: bytes=32 time=3ms TTL=128
Reply from 172.18.110.30: bytes=32 time=3ms TTL=128

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
```

## PC2

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=4ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time<1ms TTL=128
Reply from 172.18.110.62: bytes=32 time<1ms TTL=128
Reply from 172.18.110.62: bytes=32 time=3ms TTL=128
Reply from 172.18.110.62: bytes=32 time=13ms TTL=128

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 4ms
```

## PC4

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=15ms TTL=128
Reply from 172.18.110.126: bytes=32 time=3ms TTL=128
Reply from 172.18.110.126: bytes=32 time=3ms TTL=128
Reply from 172.18.110.126: bytes=32 time=5ms TTL=128

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 15ms, Average = 6ms

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time<1ms TTL=128
Reply from 172.18.110.126: bytes=32 time=5ms TTL=128
Reply from 172.18.110.126: bytes=32 time=3ms TTL=128
Reply from 172.18.110.126: bytes=32 time=3ms TTL=128

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 2ms
```

## ***REFLECTION***

What have you learned in this task?

In lab3, I learned how to configure IP addresses, subnet masks, and routing on routers and the importance of setting correct default gateways on PCs to make smooth communication across networks possible. I have come to understand, through RIP, how routers share information on routing, hence making the communication process easier between devices across different subnets.

However, the most important thing that I have learned is that even small adjustments in the router configuration may bring an entire network down because of small things such as IP addresses, therefore, a proper plan and testing have to be done. I also came to know how subnetting has the potential to optimize IP address use, and the CLI and ping are some of the very important tools in diagnosing and fixing the issues within any network.

This lab enhanced my technical skills and gave me a different perspective on how some of the routing protocols, though simple, could provide the functionality required for managing a network.